Bladder (urothelial) cancer

NMIBC (TURBT + intravesical therapy; BCG is the backbone), **MIBC** (perioperative systemic therapy + cystectomy, or trimodality preservation), and **metastatic** (**EV + pembrolizumab** first line). **Enfortumab vedotin + pembrolizumab now anchors both MIBC and metastatic disease**

Compiled from the Stanford Oncology Handbook 2025, GU: Bladder Cancer (Khaki AR, Huober M) · NCCN · primary trial reports verified against PubMed, September 2026.

WORKUP

- **Painless gross hematuria = bladder cancer until proven otherwise** (smokers, aromatic amines, cyclophosphamide, pelvic RT, aristolochic acid, schistosomiasis → squamous). Microhematuria: risk-stratified cystoscopy ± imaging
- **Cystoscopy + TURBT with detrusor muscle in the specimen** — without muscle, T2 cannot be excluded. Bimanual exam under anesthesia
- **Re-TURBT at 2–6 wk** for **any T1**, incomplete resection, or no muscle in a high-grade specimen — up to half of T1 tumors are upstaged or have residual disease
- **CT urogram** (or MR urogram / retrograde pyelogram if renal function is poor) — the **upper tracts must be imaged**: urothelium is a field defect. Urine cytology catches high-grade and CIS
- **MIBC:** CT chest (or PET/CT), CBC, CMP. **Bone scan / brain MRI only if symptomatic** (brain MRI for small cell). Image **before** TURBT when possible — post-TURBT inflammation mimics T3/N+
- **Germline testing if young** or Lynch-type family history — especially **upper tract** tumors (MSH2)

STAGING (AJCC 8TH)

Ta / Tis	Non-invasive papillary / **carcinoma in situ** (flat, high grade — the dangerous one)
T1	Invades lamina propria — **still NMIBC**, but high-risk
T2	**Muscularis propria** — a inner half, b outer half. **The NMIBC / MIBC line**
T3 / T4	Perivesical fat (a micro, b macro) / **a** prostatic stroma, SV, uterus, vagina · **b** pelvic or abdominal wall
N	N1 single true-pelvis node · N2 multiple · **N3 common iliac**
Stage	**II** T2N0 · **IIIA** T3–4aN0 or N1 · **IIIB** N2–3 · **IVA** T4b or M1a (nodes beyond common iliac) · **IVB** M1b

~75% present as NMIBC (Ta/Tis/T1) — recurrence is the rule, progression the risk. **Grade (low vs high) matters in NMIBC**; essentially all MIBC is high grade.

VARIANT HISTOLOGY

Variant	What to do
Micropapillary	Aggressive, often understaged. **Early cystectomy even for T1**; BCG responses are poor
Plasmacytoid	**CDH1** loss; spreads along fascial planes and **peritoneum** — high positive-margin rate
Sarcomatoid	Poor prognosis; treat as high-grade UC (NAC then cystectomy)
Small cell / NE	**Treat like SCLC:** neoadjuvant **etoposide + platinum**, then cystectomy or chemoRT. Brain MRI
Pure squamous / adeno	Chemo-resistant — surgery-first. Urachal adeno (dome): partial cystectomy + umbilicus; **colorectal-type regimens**

TREATMENT PATHWAY

NMIBC — Ta / Tis / T1

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TURBT (+ re-TURBT for T1) → risk group
low: single instillation, surveillance
intermediate: intravesical chemo or BCG ×1 yr
high: **BCG induction + 3 yr maintenance**
± **durvalumab** if BCG-naive (POTOMAC)

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BCG-unresponsive
radical cystectomy is standard
CIS, bladder-sparing: **TAR-200**, N-803 + BCG, nadofaragene, pembrolizumab, gem/docetaxel

MIBC — cT2–T4a N0–1

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Perioperative therapy + cystectomy
EV + pembro ×3–4 → RC → adjuvant EV + P
any cisplatin status (KN-905 vs RC; KN-B15 vs GC)
cis-eligible alternative: **GC + durvalumab** (NIAGARA) or **ddMVAC / GC** alone

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Adjuvant if no perioperative IO
pT3–4 / N+ (no NAC) or ypT2+ / ypN+:
nivolumab ×1 yr (CheckMate 274)
ctDNA+ → atezolizumab (IMvigor011, OS gain)

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Bladder preservation (selected)
maximal TURBT → **chemoRT** (5-FU/MMC, cis, gem)
solitary, <5 cm, no hydro, no extensive CIS
salvage cystectomy for persistent MIBC

Metastatic / unresectable

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First line — any platinum status
EV + pembrolizumab (EV-302, OS 31.5 vs 16.1 mo)
EV not possible: cis-fit → **nivo + GC**, or
GC / gem-carbo → **avelumab maintenance**

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Later lines — test FGFR3 and HER2
FGFR3 → erdafitinib · HER2 3+ → T-DXd
after EV+P: **platinum doublet**
taxane / clinical trial

NMIBC RISK GROUPS (AUA)

Risk	Tumor
Low	**Low-grade, solitary Ta ≤3 cm**; PUNLMP
Intermediate	Low-grade Ta that is >3 cm, multifocal, or recurs <1 yr · **high-grade Ta ≤3 cm** · low-grade T1
High	**High-grade T1** · **any CIS** · high-grade Ta >3 cm or multifocal · recurrent HG Ta · variant histology · LVI · HG prostatic urethra · BCG failure

EAU adds a **very-high-risk** tier (e.g. T1HG + CIS, multiple/large T1HG, variant histology, LVI) where **immediate cystectomy** is offered.

NMIBC MANAGEMENT

- **Single post-TURBT instillation** (**gemcitabine 2 g** or mitomycin) within 24 h for suspected low/intermediate risk — SWOG S0337: recurrence HR 0.66. **Skip if perforation** or extensive resection
- **Low risk:** surveillance cystoscopy (3 mo, then spaced out). **Intermediate:** intravesical chemo or **BCG ×1 yr**. **High: BCG induction + maintenance ×3 yr**
- **BCG:** induction weekly ×6 (start 2–4 wk post-TURBT) → **SWOG maintenance: weekly ×3 at months 3, 6, 12, 18, 24, 30, 36**. Maintenance doubled median RFS (76.8 vs 35.7 mo)
- **BCG-naive high-risk: + durvalumab** ×13 cycles (POTOMAC: DFS HR 0.68; **FDA 28 May 2026**) — grade 3–4 TRAEs 21% vs 4%
- **BCG shortage:** reserve BCG for high risk; **sequential gemcitabine + docetaxel** is the workhorse alternative (phase 2: 3-yr HG-RFS 80%); randomized BRIDGE (vs BCG) pending

BCG-UNRESPONSIVE (FDA DEFINITION)

- After **adequate BCG** (≥5/6 induction + ≥2 maintenance or second-induction doses): **persistent/recurrent CIS within 12 mo**, **HG Ta/T1 recurrence within 6 mo**, or **T1HG at the first post-induction check**
- **Radical cystectomy is the standard**; bladder-sparing options (page 2) are for patients who decline or are unfit. **Any muscle invasion → treat as MIBC**

SURVEILLANCE

- **NMIBC:** cystoscopy + cytology q3–6 mo early, then annually (low-risk can space out faster); **upper-tract imaging every 1–2 yr in high risk**
- **Post-cystectomy:** CT chest/abdomen/pelvis q3–6 mo ×2 yr, then annually; urethral wash if prostatic urethral involvement; B12 and renal function

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BCG-UNRESPONSIVE CIS — BLADDER-SPARING (ALL SINGLE-ARM)

Agent · trial	Result · status
TAR-200 (gemcitabine intravesical system) *SunRISe-1*	**CR 82.4%**, median DOR 25.8 mo; grade ≥3 TRAEs 13%. **FDA 9 Sep 2025** (Inlexzo). Indwelling device, exchanged q3wk
N-803 + BCG (nogapendekin alfa) *QUILT-3.032*	CR 71% (FDA review 62%), median DOR 26.6 mo. **FDA 22 Apr 2024** (Anktiva). Needs BCG supply
Nadofaragene firadenovec *CS-003*	Adenoviral IFN-α2b gene therapy, **q3 months**: CR 53%, 46% of responders durable at 12 mo. **FDA 16 Dec 2022**
Pembrolizumab IV *KEYNOTE-057*	CR 41% at 3 mo, but few durable. FDA Jan 2020. Systemic irAE risk for a bladder-confined disease
Gemcitabine + docetaxel	Off-label, retrospective; cheap and widely used

Cretostimogene (oncolytic adenovirus, BOND-003: CR 75%, no grade ≥3 TRAEs) is **not yet approved** (BLA expected late 2026). **Durable CR at 12 mo is the number that matters** — roughly 20–45% across agents.

CISPLATIN ELIGIBILITY (GALSKY)

- **Ineligible if any:** ECOG ≥2 · **CrCl <60 mL/min** · hearing loss grade ≥2 · neuropathy grade ≥2 · NYHA class III HF
- CrCl 50–60: **split-dose cisplatin** (35 mg/m² d1 + d8) is common practice — measured 24-h CrCl can rescue borderline cases
- **Carboplatin does not replace cisplatin in the neoadjuvant setting** — no survival benefit shown. In metastatic disease gem/carbo is acceptable

MIBC — PERIOPERATIVE THERAPY

Trial	Design · result
KEYNOTE-B15 / EV-304	**Cis-eligible:** perioperative **EV + pembro vs neoadjuvant GC** — 2-yr EFS 79.4% vs 66.2% (HR 0.53); **OS HR 0.65**; pCR 55.8% vs 32.5%. Grade ≥3 AEs 76% vs 67%. **FDA 10 Jul 2026**
KEYNOTE-905 / EV-303	**Cis-ineligible or declined:** EV + pembro vs **cystectomy alone** — 2-yr EFS 74.7% vs 39.4% (HR 0.40); **OS HR 0.50**; pCR 57.1% vs 8.6%. **FDA 21 Nov 2025**
NIAGARA	**GC + durvalumab** → RC → durva ×8 vs GC → RC — 2-yr EFS 67.8% vs 59.8% (HR 0.68), **OS HR 0.75**. FDA 28 Mar 2025
VESPER	ddMVAC ×6 vs GC ×4: neoadjuvant subgroup **5-yr OS 66% vs 57%** (HR 0.71) — **ddMVAC is the stronger chemo backbone**, at the cost of more toxicity
SWOG 8710 · ABC meta-analysis	Neoadjuvant MVAC: OS 77 vs 46 mo, pT0 38% vs 15%. Platinum combination NAC: **5% absolute 5-yr OS gain** (HR 0.87)

No head-to-head of EV + pembro vs GC + durvalumab. EV+P now has OS data against both cystectomy alone and GC. Cystectomy = bladder, prostate/SV or uterus/anterior vagina, **bilateral pelvic LND**, and diversion (conduit or neobladder).

ADJUVANT AFTER CYSTECTOMY

Trial	Result
CheckMate 274	**Nivolumab ×1 yr** vs placebo, ypT2+/ypN+ after NAC or pT3+/N+ without — DFS 20.8 vs 10.8 mo (HR 0.70). 5-yr: DFS HR 0.74, **OS 75.0 vs 50.1 mo, HR 0.83 — not yet significant** (follow-up ongoing). Benefit concentrated in PD-L1 ≥1%
IMvigor011	Serial ctDNA ×1 yr; **ctDNA+ → atezolizumab vs placebo**: DFS 9.9 vs 4.8 mo (HR 0.64), **OS 32.8 vs 21.1 mo (HR 0.59)**. Persistently ctDNA–: 2-yr DFS 88% untreated. **FDA 15 May 2026** (Signatera CDx)
AMBASSADOR	Pembrolizumab ×1 yr vs observation — DFS 29.6 vs 14.2 mo (HR 0.73); no OS gain shown

METASTATIC — FIRST LINE

Regimen · trial	Result
EV + pembrolizumab *EV-302*	vs platinum-gem, cis- and carbo-eligible alike: **OS 31.5 vs 16.1 mo (HR 0.47)**, PFS 12.5 vs 6.3 mo (HR 0.45). Grade ≥3 TRAEs 56% vs 70%. **The standard for everyone fit for it; no biomarker needed**
Nivo + gem/cis *CheckMate 901*	Cis-eligible: OS 21.7 vs 18.9 mo (HR 0.78), PFS HR 0.72, CR 21.7% vs 11.8% with durable CRs. **The option when EV is contraindicated** (neuropathy, uncontrolled diabetes)
Platinum → avelumab *JAVELIN Bladder 100*	No progression after 4–6 cycles gem/cis or gem/carbo: **avelumab maintenance** — OS 21.4 vs 14.3 mo (HR 0.69). Start within 4–10 wk of the last cycle
Pembrolizumab alone	**Only if platinum-ineligible** and EV-ineligible (label restricted 2021)

Don't give chemo + pembrolizumab or chemo + atezolizumab (KEYNOTE-361, IMvigor130: no OS gain) — **nivolumab is the only checkpoint inhibitor that adds OS to 1L platinum**.

BIOMARKERS & LATER LINES

Target / setting	Agent · evidence
FGFR3 alteration *~15–20% of mUC*	**Erdafitinib** after ≥1 line incl. IO (THOR: **OS 12.1 vs 7.8 mo** vs chemo, HR 0.64). Full approval Jan 2024, **FGFR3 only** (FGFR2 dropped). **Not instead of IO** in IO-naive
HER2 IHC 3+	**T-DXd 5.4 mg/kg** — tumor-agnostic accelerated approval (Apr 2024; DESTINY-PanTumor02 IHC 3+ ORR 61%). ILD 10.5%
HER2 IHC 1–3+, 1L	**Disitamab vedotin + toripalimab** vs chemo: OS 31.5 vs 16.9 mo (HR 0.54). **Approved in China, not US**
After EV + pembro	**Platinum doublet** (gem/cis or gem/carbo); FGFR3 / HER2 agents; taxane; trial
After platinum ± IO, EV-naive	**EV monotherapy** (EV-301: OS 12.9 vs 9.0 mo, HR 0.70). IO-naive after platinum: pembrolizumab (KEYNOTE-045, OS 10.3 vs 7.4 mo)
Sacituzumab govitecan	**Withdrawn for UC (Oct 2024)** — TROPiCS-04 missed OS (HR 0.86) with **7% grade 5 AEs**, mostly neutropenic infection

TRIMODALITY THERAPY (BLADDER PRESERVATION)

- **Maximal TURBT + concurrent chemoRT**, then cystoscopic reassessment; **salvage cystectomy** for persistent/recurrent MIBC
- **Best candidates:** solitary T2, <5 cm, visibly complete TURBT, **no hydronephrosis**, no extensive CIS, good bladder capacity
- **BC2001:** RT + **5-FU/mitomycin** vs RT alone — 2-yr locoregional DFS 67% vs 54% (HR 0.68); OS 48% vs 35% (NS). No randomized comparison with cystectomy
- **Lifelong cystoscopic surveillance** — NMIBC recurrences in the preserved bladder are treated with TURBT ± BCG

UPPER TRACT UROTHELIAL CARCINOMA

- **~5–10% of UC**; renal pelvis > ureter. **Lynch syndrome** (MSH2) — screen with MMR IHC / germline testing; aristolochic acid (Balkan nephropathy)
- Diagnosis: CT urogram + **ureteroscopy with biopsy** and cytology. Low-grade, small: **kidney-sparing** (endoscopic ablation, mitomycin gel)
- High-risk: **radical nephroureterectomy with bladder cuff**. **Consider neoadjuvant chemo** — GFR falls after NU and may exclude cisplatin later
- **POUT:** adjuvant gem + platinum (carboplatin if GFR <50) after NU for pT2–4/N+ — DFS HR 0.45; 5-yr OS 66% vs 57% (HR 0.68)
- **20–50% later develop bladder tumors** — cystoscopic surveillance

Bladder (urothelial) cancer

REGIMEN DOSING

Regimen	Dosing
BCG (TICE) intravesical	1 vial (50 mg) in 50 mL saline, dwell 2 h. **Induction weekly ×6**; maintenance **weekly ×3 at mo 3, 6, 12, 18, 24, 30, 36**
Durvalumab + BCG	Durvalumab **1500 mg IV q4wk ×13** + BCG induction + maintenance
Post-TURBT instillation	**Gemcitabine 2 g** in 100 mL (dwell 1 h) or mitomycin 40 mg, <24 h
Gem / docetaxel intravesical	Gemcitabine 1 g, drain, then docetaxel 37.5 mg (2 h dwell) weekly ×6 → monthly maintenance
TAR-200	Gemcitabine 225 mg device, **q3wk ×8 (6 mo) → q12wk ×6**; removed after each 3-wk dwell
N-803 + BCG	N-803 **400 µg** + BCG weekly ×6 (repeat once if no CR at 3 mo); maintenance ×3 wk at mo 4, 7, 10, 13, 19
Nadofaragene	75 mL (3×10^{11} vp/mL) intravesical **q3 months**
ddMVAC	MTX 30, vinblastine 3, doxorubicin 30, **cisplatin 70** mg/m² d1 **q14d** + pegfilgrastim; **×3–6 neoadjuvant**
Gem / cis	Cisplatin **70** d1 + gemcitabine **1000** d1, 8 **q21d** (×4 neoadjuvant; ≤6 metastatic). Split: cis 35 d1 + d8
+ Durvalumab (NIAGARA)	**1500 mg** d1 with GC ×4 → RC → 1500 mg q4wk ×8
EV + pembro, metastatic	EV **1.25 mg/kg** (max 125 mg) **d1, 8** + pembro 200 mg d1, **q21d**; pembro max 2 yr
EV + pembro, perioperative	Cis-eligible: **×4 neoadj** → RC → EV ×5 + pembro ×13. Cis-ineligible: **×3 neoadj** → RC → EV ×6 + pembro ×14
Nivo + gem/cis	Nivolumab **360 mg** + GC q3wk ×≤6 → nivolumab 480 mg q4wk, max 2 yr
Gem / carbo	Carboplatin AUC 4.5–5 d1 + gemcitabine 1000 d1, 8 q21d ×4–6
Avelumab maintenance	**800 mg IV q2wk** until progression
Adjuvant IO	Nivolumab 240 q2wk / 480 q4wk ×1 yr · atezolizumab 1680 q4wk ×1 yr (ctDNA+)
Erdafitinib	**8 mg PO daily**; check phosphate at d14–21 → **9 mg if PO_4 <9.0 mg/dL** and no eye toxicity
EV monotherapy	1.25 mg/kg d1, 8, 15 q28d
ChemoRT (BC2001)	5-FU 500 mg/m²/d fractions 1–5 + 16–20, **mitomycin 12 mg/m²** d1; RT 55 Gy/20 fx or 64 Gy/32 fx. Alt: weekly cisplatin 35–40 or twice-weekly gemcitabine 27 mg/m²

TOXICITY & MONITORING

- **Enfortumab vedotin — boxed warning: SJS/TEN**. Rash is common, usually in cycle 1: hold for grade 3 or suspected SJS/TEN, **permanently stop if confirmed**. Also **peripheral neuropathy** (cumulative, most common reason to stop), **hyperglycemia/DKA** (trials excluded HbA1c ≥8%), dry eye/keratitis, pneumonitis, extravasation
- **EV + pembro:** rash can be EV or immune — a derm consult early saves the regimen. Dose-reduce EV (1.0 → 0.75 → 0.5 mg/kg) rather than dropping it
- **Erdafitinib: hyperphosphatemia** is on-target (FGFR blocks FGF23 signaling) — low-phosphate diet, binders, dose adjust. **Central serous retinopathy**: ophthalmology exam **monthly ×4 mo, then q3 months**, and urgently for visual change. Also nail/hand-foot changes, stomatitis, dry eye
- **BCG:** cystitis, fever <48 h is common. **Do not instill with traumatic catheterization, gross hematuria, or active UTI** (intravasation risk). **BCG sepsis / BCGitis** (persistent fever, pneumonitis, hepatitis, mycotic aneurysm): **isoniazid + rifampin + ethambutol** ± steroids — **M. bovis is intrinsically pyrazinamide-resistant**. Handle as a biohazard
- **Cisplatin:** nephrotoxicity (pre/post hydration, Mg), ototoxicity, neuropathy; **re-check eligibility every cycle**. ddMVAC: neutropenia (G-CSF mandatory), mucositis, anthracycline cardiotoxicity
- **Checkpoint inhibitors:** standard irAE surveillance — TSH, LFTs, cortisol; watch for pneumonitis and colitis. Intravesical agents (TAR-200, N-803) are mostly local urinary toxicity
- **After cystectomy:** ileal conduit/neobladder → **hyperchloremic metabolic acidosis**, B12 deficiency, stones, UTIs; neobladder needs a nocturnal voiding schedule

PEARLS

- **No muscle in the TURBT specimen = not staged**
- **EV + pembro is now the answer in three settings**: cis-ineligible MIBC, cis-eligible MIBC, and 1L metastatic
- **Nivolumab is the checkpoint inhibitor that works with platinum**; pembro and atezo did not

KEY TRIALS

EV-302	1L metastatic EV + pembro — OS HR 0.47
KEYNOTE-B15 / EV-304	Perioperative EV + P vs GC, cis-eligible — OS HR 0.65
KEYNOTE-905 / EV-303	Perioperative EV + P vs RC alone — OS HR 0.50
NIAGARA	GC + perioperative durvalumab — OS HR 0.75
VESPER	Neoadjuvant ddMVAC > GC (5-yr OS 66 vs 57%)
SWOG 8710	Neoadjuvant MVAC — OS 77 vs 46 mo
CheckMate 274	Adjuvant nivolumab — DFS HR 0.70
IMvigor011	ctDNA-guided adjuvant atezolizumab — OS HR 0.59
CheckMate 901	Nivo + gem/cis 1L — OS HR 0.78
JAVELIN Bladder 100	Avelumab maintenance — OS HR 0.69
THOR	Erdafitinib, FGFR3 after IO — OS HR 0.64
EV-301	EV after platinum + IO — OS HR 0.70
SWOG 8507	BCG maintenance schedule — RFS 76.8 vs 35.7 mo
POTOMAC	Durvalumab + BCG, BCG-naive — DFS HR 0.68
SunRISe-1	TAR-200, BCG-unresponsive CIS — CR 82%
BC2001	ChemoRT (5-FU/MMC) > RT — LRDFS HR 0.68
POUT	Adjuvant chemo for UTUC — DFS HR 0.45